

Customer Shopping Behavior Analysis

Insights from **3,900 transactions** across product categories, demographics, and shopping patterns.



Dataset at a Glance

3,900

Purchases

18

Columns

37

Missing Values

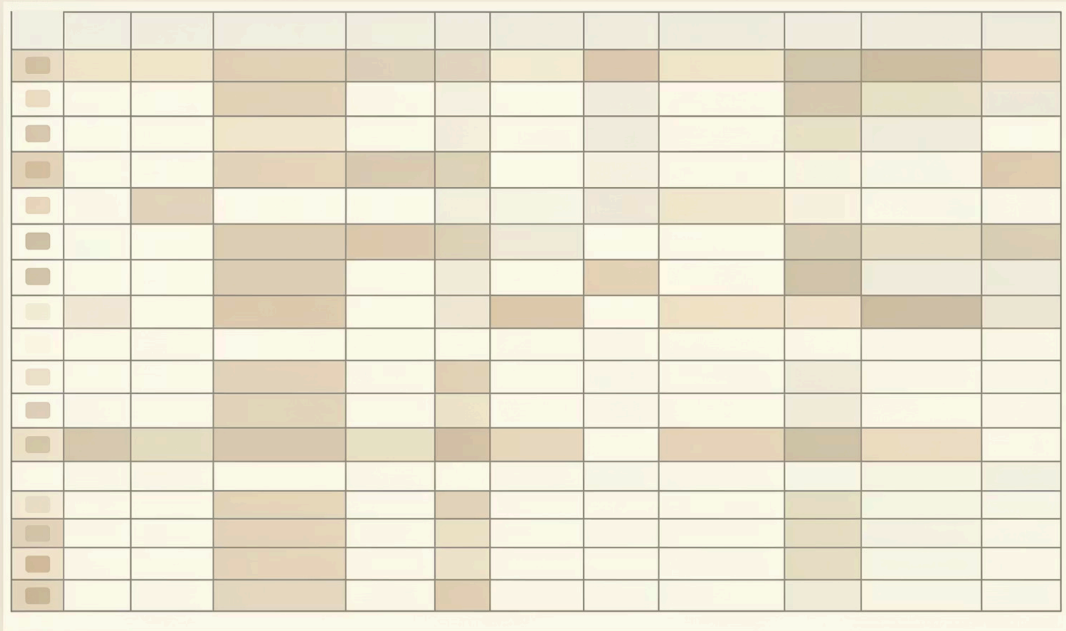
Review Rating only

\$59.76

Avg. Purchase

Key Features

- Demographics: Age, Gender, Location, Subscription
- Purchase details: Category, Amount, Season, Size, Color
- Behavior: Discount, Frequency, Review Rating, Shipping



Data Preparation in Python

Load & Explore

1

Imported via pandas; checked structure with `df.info()` and `.describe()`

Clean & Standardize

2

Imputed missing Review Ratings by category median; renamed columns to snake_case

Feature Engineering

3

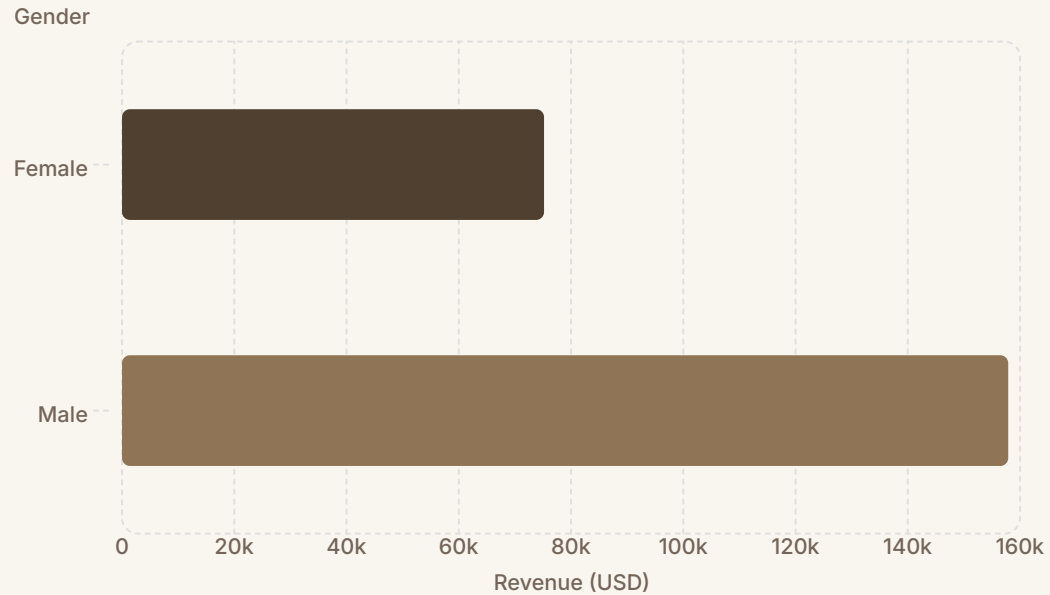
Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used`

Database Integration

4

Loaded cleaned DataFrame into PostgreSQL for SQL analysis

Revenue by Gender



Key Finding

Male customers generated **2.1× more revenue** than female customers across 3,900 purchases.

Male: \$157,890 · Female: \$75,191

Top Products & Discount Patterns

Top 5 Products by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Shipping & Subscription Insights

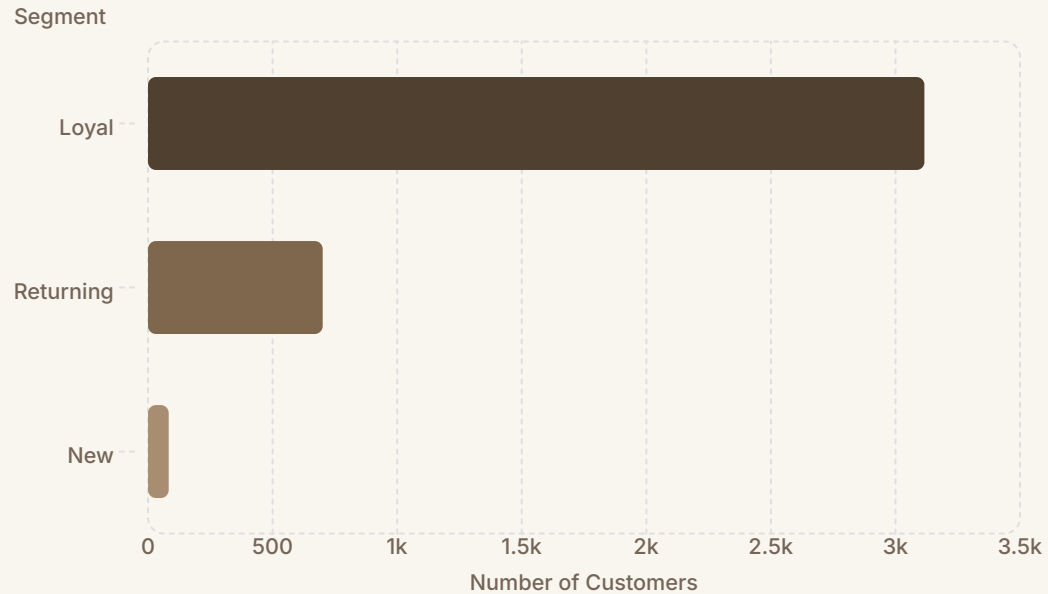
Shipping Type — Avg. Purchase

Type	Avg Spend
Standard	\$58.46
Express	\$60.48

Subscribers vs. Non-Subscribers

Status	Customers	Avg Spend	Revenue
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

Customer Segmentation



Segment Definitions

Loyal

3,116 customers — highest purchase frequency

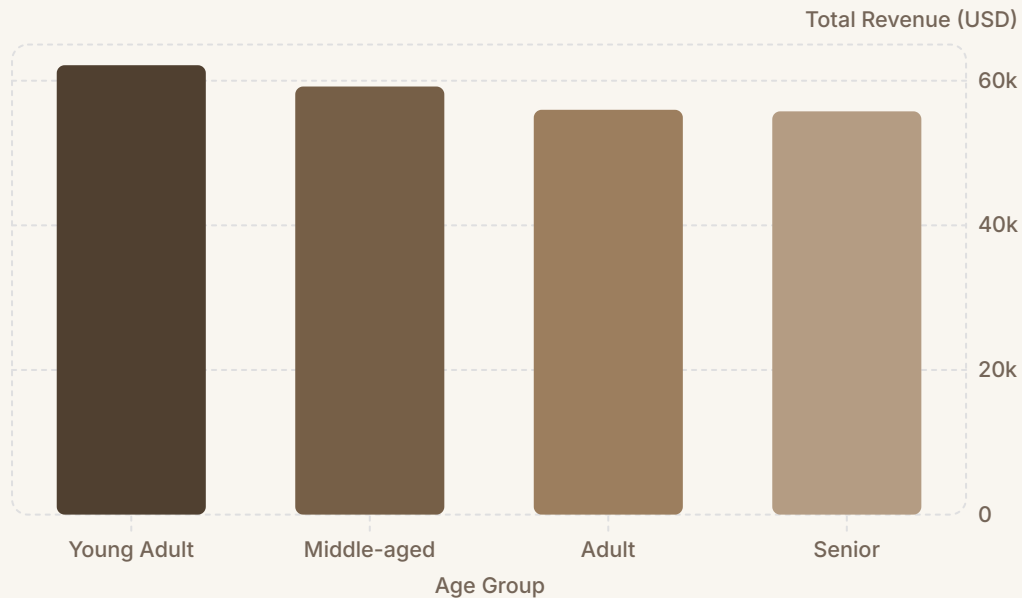
Returning

701 customers — repeat buyers

New

83 customers — first-time purchasers

Revenue by Age Group



Key Insight

Young Adults lead in revenue contribution at **\$62,143**, though all age groups are relatively close — indicating broad market reach.

i Repeat buyers with 5+ purchases: 958 subscribed vs. 2,518 not subscribed

Power BI Dashboard



Dashboard Highlights

KPIs

3.9K customers · \$59.76 avg purchase · 3.75 avg rating

Filters

Subscription, Gender, Category, Shipping Type

Charts

Revenue & Sales by Category and Age Group

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow Loyal segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users



Product Positioning

Highlight top-rated & best-selling products in campaigns